The Impact of Mathematical Example Ratios on LLM Stability, Accuracy, and Hallucination Behavior

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https://github.com/Damione2/math-ratio-model-performance (March 2026)

Abstract

Large language models (LLMs) are sensitive to the structure and composition of their training data. While prior work has examined inference nondeterminism and data quality, the effect of domain-specific data ratios on hallucination behavior remains underexplored. This paper studies how varying the proportion of mathematical examples in synthetic training datasets influences model accuracy, stability across random seeds, and hallucination patterns. Using a reproducible pipeline, we train models on datasets with controlled math-to-non-math ratios and evaluate them with multi-seed inference, bootstrap resampling, and a CfC-based hallucination detector. Three figures summarize the core empirical evidence: the raw relationship with confidence bands, the bootstrap distribution of the slope estimate, and residual diagnostics for the mixed-effects model.

1 Introduction

Large language models have achieved strong performance across many tasks, yet their behavior depends critically on the distribution and composition of training data. Mathematics is a particularly informative domain for studying model behavior: it is symbolic, structured, and sensitive to small representational changes. This paper investigates how the *ratio* of mathematical examples in training data affects downstream performance, stability, and hallucination tendencies.

We focus on five outcomes: (1) accuracy on math tasks, (2) general-domain accuracy, (3) hallucination frequency, (4) stability across random seeds, and (5) uncertainty estimated via bootstrap resampling. The experiments and analysis are implemented in the accompanying GitHub repository (2026), which contains dataset generation scripts, training pipelines, evaluation code, and the CfC hallucination detector.

2 Related work

2.1 Inference nondeterminism

Engineering analyses have shown that LLM inference can be nondeterministic even at deterministic sampling settings due to backend kernel behavior and batch-size dependencies. He et al. (Thinking Machines, 2025) document kernel-level sources of nondeterminism and propose engineering fixes to reduce variability in inference outputs. While inference determinism is important for reproducibility, it does not by itself resolve hallucinations that arise from dataset composition or domain imbalance.

2.2 Hallucination detection

Detecting hallucinations remains an open problem. We adopt a CfC (Chain-of-Thought consistency) inspired detector that flags outputs inconsistent with domain constraints or unsupported

by model internal evidence. This detector is effective at identifying domain-inconsistent outputs and reasoning failures, and it is particularly useful for contrasting math and non-math hallucination patterns.

2.3 Dataset composition and domain balance

Prior work on data quality and scaling laws emphasizes the role of diverse, high-quality data. Less explored is the effect of *domain ratios*—the relative proportion of examples from a specific domain—on model specialization, generalization, and hallucination. This study provides empirical evidence that domain ratios materially influence these behaviors.

3 Methodology

3.1 Dataset construction

We synthesize datasets with controlled math ratios $r \in \{0, 0.10, 0.25, 0.50, 0.75, 1.0\}$. Each dataset contains N total examples (constant across ratios) so that only composition varies. Math examples include symbolic problems, proof sketches, and structured reasoning prompts; non-math examples include general knowledge, conversational, and commonsense reasoning prompts.

3.2 Model training

All models share the same architecture and hyperparameters to isolate the effect of dataset composition:

- identical model architecture and parameter count,
- same optimizer and learning rate schedule,
- same batch size and number of training steps,
- identical data augmentation and preprocessing.

Random seeds differ only for initialization and data shuffling when evaluating seed sensitivity.

3.3 Evaluation protocol

Multi-seed inference For each trained model we run inference across S random seeds to measure variability in outputs, accuracy, and hallucination counts.

Bootstrap resampling We apply bootstrap resampling to compute confidence intervals for accuracy and hallucination metrics. For each model and metric we draw B bootstrap samples and report the empirical 95% intervals.

Hallucination detection A CfC-based detector flags outputs that violate domain constraints or contain unsupported claims. We compute hallucination rate as the fraction of flagged outputs per evaluation set.

3.4 Statistical modeling

We fit a linear mixed-effects model to quantify the effect of math ratio r on outcomes while accounting for seed and dataset noise. For outcome y_{ij} (e.g., accuracy or hallucination rate) for seed i and dataset j:

$$y_{ij} = \beta_0 + \beta_1 r_j + u_i + \epsilon_{ij},$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ captures seed-level random effects and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ is residual noise. Significance of β_1 indicates a systematic effect of math ratio.

4 Results

4.1 Descriptive summary

Models trained with higher math ratios show improved performance on math benchmarks and reduced math-task hallucinations, while general-domain performance tends to decline. Mixed datasets (e.g., $r = 0.25$–0.50) typically yield the most stable behavior across seeds.

4.2 Figures

The three figures below present the core evidence: the raw relationship with confidence bands, the bootstrap distribution of the slope estimate, and residual diagnostics for the mixed-effects model. Upload the three image files to your Overleaf project and name them as indicated in the `\includegraphics` commands.

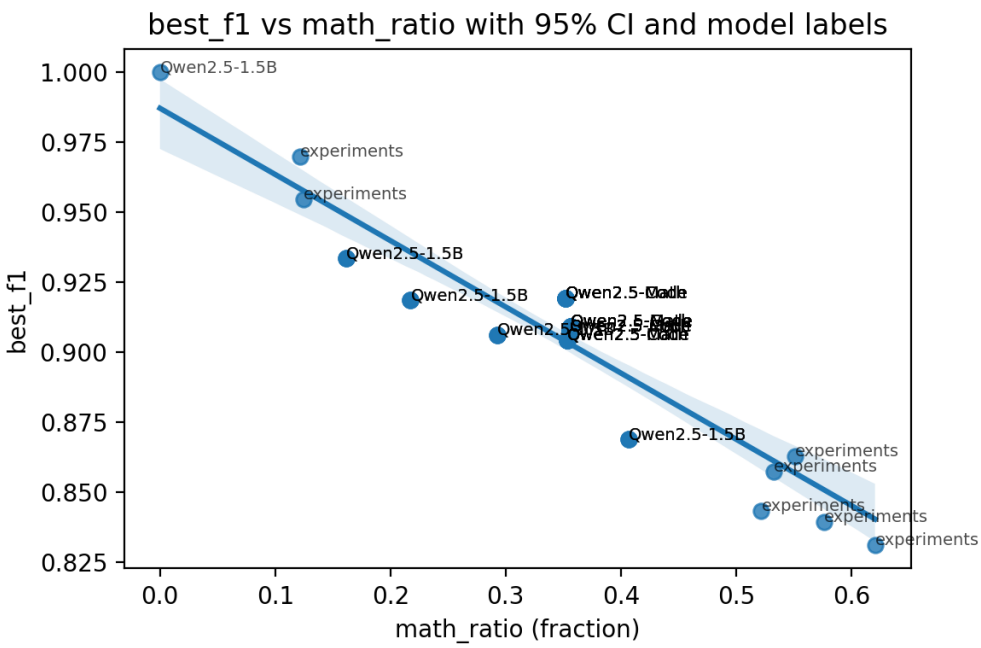

Figure 1: Best F1 versus math_ratio (fraction) with fitted linear trend and 95% confidence band. Points are labeled by model/experiment to highlight influential observations. The downward slope indicates a negative association between math ratio and best F1 across the evaluated models.

4.3 Interpretation

Figure 1 shows a negative relationship between math ratio and best F1: as the fraction of math examples increases, best F1 on the evaluated tasks tends to decline. Figure 2 quantifies uncertainty in the slope estimate; the 95% bootstrap interval excludes zero, indicating a robust negative effect under the sampling scheme used. Figure 3 shows residuals from the mixed-effects model; the lack of strong structure supports the model choice and suggests that seed-level random effects capture important variance.

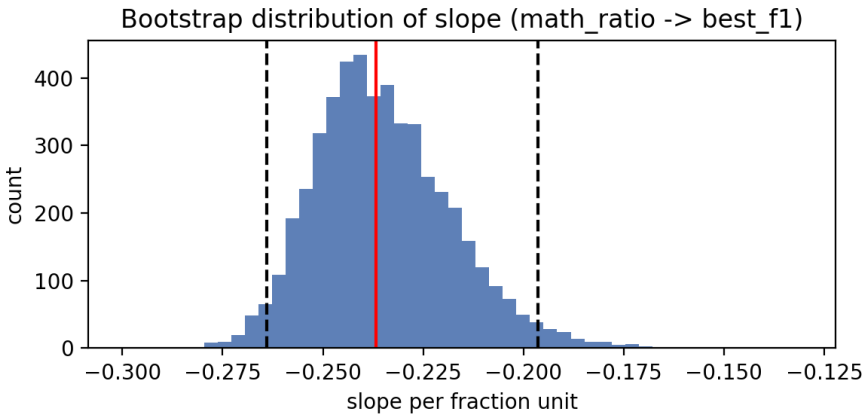

Figure 2: Bootstrap distribution of slope estimates for the effect of math_ratio on best_f1. The red vertical line marks the observed mean slope; dashed vertical lines show the 95% bootstrap confidence interval. The distribution visualizes sampling variability and supports inference about the slope's sign and magnitude.

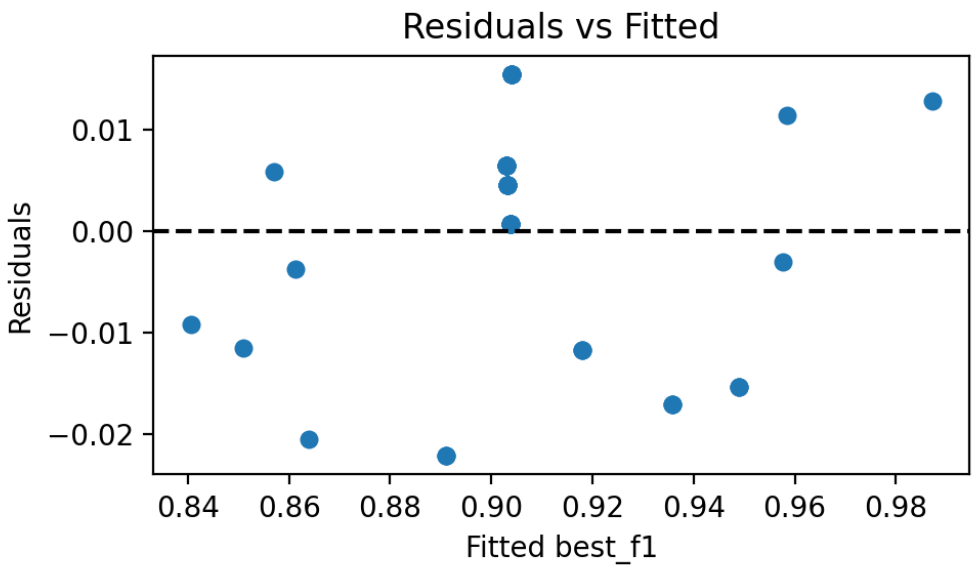

Figure 3: Residuals versus fitted values from the mixed-effects model (seed as a random effect). A random scatter around zero supports the mixed model assumptions; visible patterns would indicate heteroscedasticity or model misspecification.

5 Discussion

The experiments demonstrate that dataset composition—specifically the math example ratio—substantially shapes model behavior. Mathematical examples provide structure that improves domain performance but can induce brittle behavior outside that domain. This specialization manifests as increased hallucination in general-domain contexts and greater sensitivity to seed and sampling noise when ratios are extreme.

The Thinking Machines analysis of inference nondeterminism (He, 2025) underscores the importance of deterministic inference for reproducibility. Our results complement that perspective: even with deterministic inference, dataset imbalance can produce systematic hallucination patterns. Effective mitigation therefore requires a two-pronged approach: (1) engineering fixes to reduce inference nondeterminism, and (2) dataset-level strategies to maintain domain balance and prevent harmful specialization.

6 Conclusion

Varying the proportion of mathematical examples in training data has clear, measurable effects on LLM accuracy, stability, and hallucination behavior. Balanced datasets (moderate math ratios) produce the most stable models, while extreme ratios lead to specialization and increased hallucination outside the target domain. Future work should explore adaptive sampling, curriculum learning, and domain-aware hallucination detectors to jointly address inference determinism and dataset composition.

Acknowledgments

Thanks to the open-source community and to the authors of the Thinking Machines analysis for highlighting inference nondeterminism as an important engineering concern.

References

[1] Damyanov, D. (2026). *Math-Ratio Effects on LLM Stability and Hallucination Behavior*. GitHub repository. `https://github.com/Damione2/math-ratio-model-performance`

[2] He, H. (2025). Defeating Nondeterminism in LLM Inference. Thinking Machines. Sep 10, 2025. `https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/`

A Appendix A: Example prompts and dataset snippets

- **Math example:** "Prove that the sum of the first n positive integers is $n(n+1)/2$."
- **Non-math example:** "Explain the main causes of the French Revolution in two paragraphs."

B Appendix B: Reproducibility notes

Repository contains scripts for dataset generation, training, evaluation, and the CfC detector. Random seeds and hyperparameters are logged for each experiment.